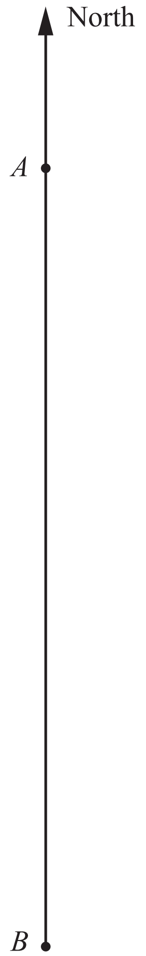
1 **(a)** The scale diagram shows the positions, *A* and *B*, of two boats.



Scale: 1 cm to 50 m

(i) Find the actual distance between the two boats.

Answer ....m [1]

(ii) A third boat is positioned at *C*, such that *AC* = 350 m and *BC* = 300 m.
C is east of the line *AB*.

Use ruler and compasses to find *C*. [2]

(iii) Measure the bearing of *C* from *A*.

Answer .... [1]

(iv) A fourth boat is positioned at *D*, such that *AC* is the line of symmetry of the quadrilateral *ABCD*.

Complete the quadrilateral *ABCD*. [2]

2 *ABCD* is a field in the shape of a trapezium.
$A\hat{B}C = 56°$, $B\hat{A}D = 104°$ and the distance between the parallel sides of the field is 90 m.

(a) Using a scale of 1 cm to 20 m, draw a plan of the field.
AB has been drawn for you.

[4]

(b) Find the actual distance *CD*.

Answer $CD =$ .... m [2]

3 In triangle ABC, $AB = 5$ cm and $AC = 6$ cm.

(a) Construct triangle ABC.
Line BC is drawn for you.

B ———————————————————— C

[2]

(b) Measure $B\hat{A}C$ in your triangle.

Answer .... [1]

4 The diagram shows the positions of three ships A, B and C.
It is drawn to a scale of 1 cm to 20 km.

(a) Find, by measurement, the bearing of C from A.

Answer .... [1]

(b) On the diagram construct the locus of points, inside triangle ABC, that are

(i) equidistant from B and C, [1]

(ii) equidistant from AB and BC. [1]

(c) A ship D is

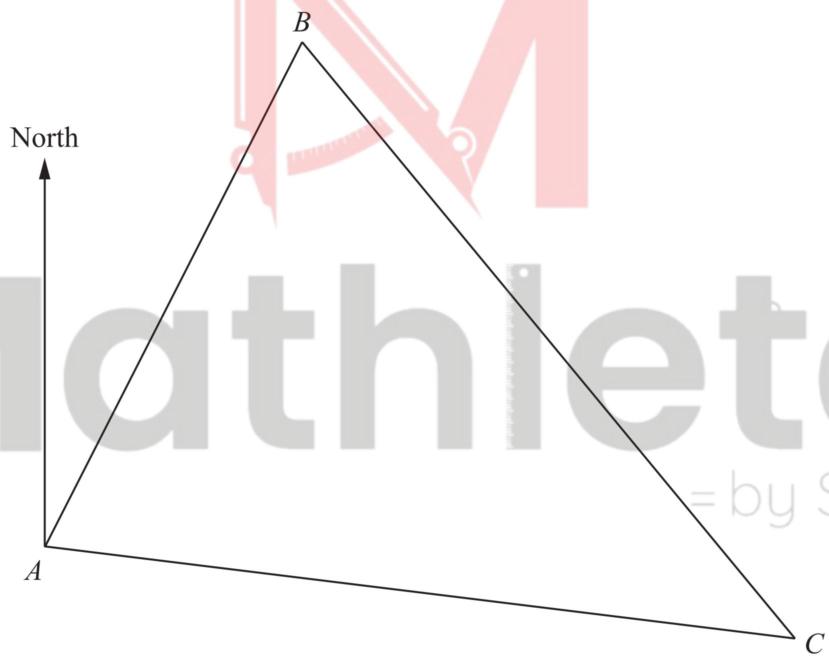
- equidistant from B and C,

and

- equidistant from AB and BC.

Label the position of D on the diagram and find the actual distance of D from A.

Scale: 1 cm to 20 km



Answer DA = ... km [1]

5 **(a)** Triangle ABC has sides $AB = 8$ cm, $AC = 7$ cm and $BC = 12$ cm.

(i) Use a ruler and compasses to construct triangle ABC.
Side AB has been drawn for you.

[2]

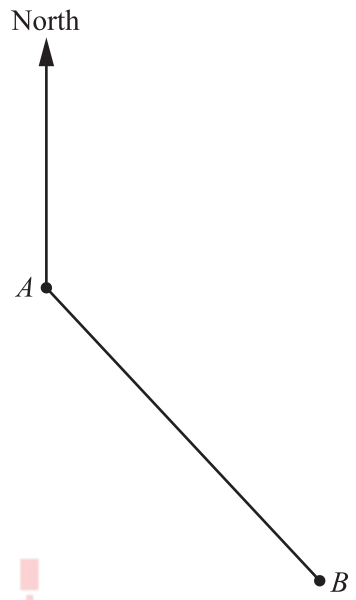
(ii) Measure $B\hat{A}C$.

Answer .... [1]

(b) Calculate the interior angle of a regular 12-sided polygon.

Answer .... [2]

6 The diagram shows the position of two villages A and B.



(a) Measure the bearing of B from A.

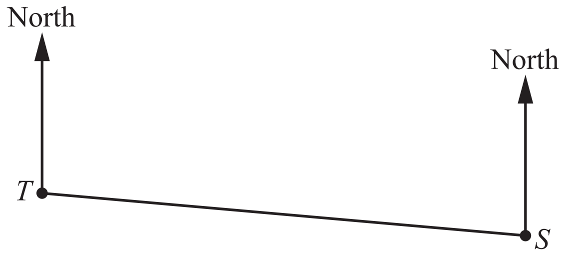
Answer .... [1]

(b) The bearing of village C from A is 265°.

Work out the bearing of A from C.

Answer .... [1]

7



The diagram shows the position of a clock tower, T, and a statue, S, drawn to a scale of 1 cm to 75 m.

(i) Using measurements taken from the diagram, find the actual distance between T and S.

Answer .... m [2]

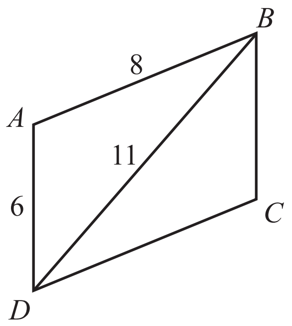
(ii) A fountain, F, is situated 450 m from T on a bearing of 210°.

Draw and label F. [2]

(iii) Using measurements taken from the diagram, find the bearing of F from S.

Answer .... [1]

8 **(a)**



NOT TO SCALE

ABCD is a parallelogram.
$AD = 6$ cm, $AB = 8$ cm and $BD = 11$ cm.

(i) **Using a ruler and compasses only**, construct an accurate drawing of *ABCD*.
AD has been drawn for you.

[3]

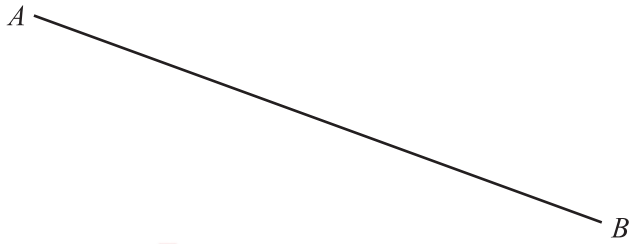
(ii) Measure $D\hat{A}B$.

$D\hat{A}B =$ .... [1]

(iii) *E* is the point on *BD* such that *AE* is the shortest distance from *A* to *BD*.

Draw and measure *AE*.

$AE =$ .... cm [1]



(a) C is the point **above** AB, where $AC = 5\,\text{cm}$ and $BC = 7\,\text{cm}$.

Using a pair of compasses and ruler only, construct triangle ABC. [2]

(b) D is the point **below** AB, where $B\hat{A}D = 28°$ and $A\hat{B}D = 96°$.

Using a protractor and ruler, draw triangle ABD. [2]

10 **(a)** In triangle PQR, $PR = 7.5$ cm and $QR = 6$ cm.

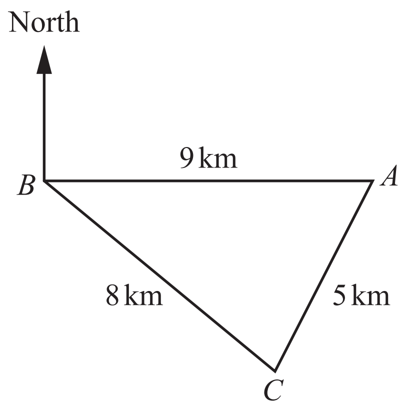
(i) **Using a ruler and compasses only**, construct triangle PQR.
Line PQ has been drawn for you.

[2]

(ii) By taking suitable measurements from your triangle, calculate the area of triangle PQR.

.... cm^2 [2]

11



NOT TO SCALE

The sketch shows the positions of three villages, A, B and C.
A is due east of B.

(a) **Use a ruler and compasses only** to complete the scale drawing of triangle ABC.
Use a scale of 1 cm to represent 1 km.

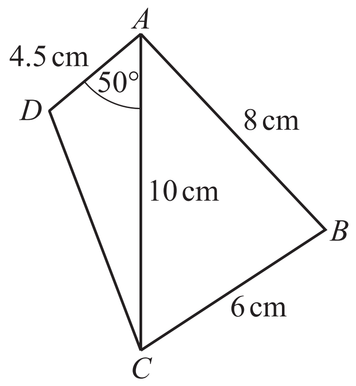
Scale: 1 cm to 1 km

[2]

(b) Measure the bearing of C from B.

.... [1]

12 **(a)** The diagram shows a sketch of quadrilateral *ABCD*.



NOT TO SCALE

(i) Construct an accurate drawing of *ABCD*.
AC has been drawn for you.

[3]

(ii) Measure $A\hat{D}C$.

.... [1]

(iii) By taking a suitable measurement from your diagram, find the perimeter of quadrilateral *ABCD*.

.... cm [1]

Answer Key:

1. **S13/qp21/q11(a)**

11 (a) (i)	510 – 520 m	1	
(ii)	C positioned 7 cm from A and 6 cm from B with both construction arcs drawn	2	**B1** for c positioned 7 cm from A and/or 6 cm from B
(iii)	146° ±2	1 ft	
(iv)	D positioned 10.3 cm ±0.8 from A and $D\hat{A}C = 34° \pm 2°$	2	**B1** for DAC = 34 (±2°)

2. **W14/qp22/q6**

6 (a)	Correct shape $ABCD$	4	**B1** for $A\hat{B}C = 56$ **B1** for $B\hat{A}D = 104$ **M1** line CD // AB **A1** for perpendicular length 4.5

3. **S15/qp12/q14**

14 (a)	Correct triangle with arcs	2	**B1** for correct triangle with no arcs or 1 arc After B0 allow **SC1** for triangle with arcs with 5 cm and 6 cm reversed
(b)	128 to 133°	1	

4. **W15/qp12/q21**

21 (a)	(0)96 to (0)98	1	
(b) (i)	Perpendicular bisector of BC.	1	
(ii)	Bisector of angle ABC.	1	
(c)	DA = 80 to 84 km	1	Dependent on two acceptable intersecting loci

5. **S16/qp22/q4(a)**

4 (a) (i)	Correct triangle with arcs shown	2	**B1** for correct triangle with no arcs or triangle with one side correct length with arcs or triangle with BC = 7 and AC = 12 with arcs (reflection)
(ii)	104 to 108	1	

6. **S17/qp12/q5**

5(a)	137	1	
5(b)	085	1	

7. S17/qp21/q6(b)

6(b)(i)	472 to 488	**2**	**B1** for 6.3 to 6.5 seen
6(b)(ii)	*F* correctly placed	**2**	**M1** for either *TF* = 6 cm plotted or correct angle
6(b)(iii)	242° to 248°	**1**	

8. S19/qp21/q5(a)

5(a)(i)	Correct parallelogram with construction arcs at *B* and *C*	**3**	**B2** for correct parallelogram without arcs **or** **B1** for two of *AB* = 8 or *DC* = 8 **or** *BD* = 11 or *BC* = 6
5(a)(ii)	101 to 105	**1**	FT *their* triangle *ABD*
5(a)(iii)	4[.0] to 4.4 with A*E* drawn	**1**	FT *theirAE* from correct parallelogram

9. W19/qp12/q19

19(a)	Acceptable triangle *ABC* with intersecting arcs at *C*	**2**	**B1** for an acceptable *C* clearly indicated with no/incorrect arcs If 0 scored, **SC1** for correct triangle *ABC* with arcs drawn below the line
19(b)	Acceptable triangle *ABD*	**2**	**B1** for one correct angle drawn If 0 scored, **SC1** for correct triangle *ABD* drawn above the line

10. W20/qp21/q3(a)

3(a)(i)	Correct triangle with intersecting arcs	**2**	**B1** for acceptable vertex *R* indicated with no/incorrect arcs After 0 scored, **SC1** for triangle with *P* = 6 cm and *QR* = 7.5 cm with intersecting arcs
3(a)(ii)	20.8 to 21.6	**2**	**M1** for $\frac{1}{2}\times 8\times \textit{their}\text{ height}$ oe or $\frac{1}{2}\times 8\times 7.5\times \sin \textit{their } 45$ or $\frac{1}{2}\times 8\times 6\times \sin \textit{their } 63$ or $\frac{1}{2}\times 6\times 7.5\times \sin \textit{their } 72$

11. S21/qp12/q9

9(a)	Acceptable triangle with intersecting arcs	**2**	**B1** for acceptable *C* clearly indicated with no/incorrect arcs
9(b)	90 + *their* angle *ABC*	**1**	**FT** *their* triangle

12. S21/qp21/q4(a)

4(a)(i)	Correct quadrilateral with construction arcs	**3**	**B2** for triangle *ABC* correct with construction arcs or triangle *ADC* correct or **B1** for *ABC* correct with no/incorrect arcs or $D\hat{A}C = 50°$
4(a)(ii)	*their* $A\hat{D}C$	**1**	FT *their* diagram
4(a)(iii)	18.5 + *their DC*	**1**	FT *their* diagram